

TECHNICAL DOCUMENTATION: DISTILLATION COLUMN SYSTEM

IDENTITY INFORMATION

****Company_Name:** Meridian Petroleum Operations LLC**

****Field_Engineer:** James R. Hutchinson**

****Field_Engineer_Id:** FE-2847-MP**

****Document_Version:** 3.2**

****Machine_Name:** Atmospheric Distillation Unit (ADU-7)**

****Machine_Id:** ADU-7-MERID-2024**

****Installation_Date:** 2023-06-15**

****Number_Of_Units_In_Machine:** 4**

UNIT 1: PRIMARY FRACTIONATION COLUMN

****Unit_Name:** Main Distillation Tower**

****Unit_Id:** DC-FRAC-001**

****Manufacture_Date:** 2023-04-22**

****Manufacturer_Company_Name:** Thermal Separation Industries Inc.**

****Manufacturer_Company_Id:** TSI-MFG-4521**

****Inventory_Quantity:** 1**

****Supplier_Information:** TSI Direct Supply, Houston TX; Contact: procurement@tsi-direct.com; Lead Time: 18 weeks**

DOMAIN SPECIFICATIONS

- ****Wellbore_Depth_Rating:** N/A (Refinery Equipment)**
- ****Operating_Pressure_PSI:** 35–42 PSI (atmospheric + slight vacuum)**
- ****Drilling_Fluid_Compatibility:** N/A (Crude oil feedstock: API gravity 28–35°)**
- ****Known_Error_Codes:** ERR-DC-001 (Tray flooding), ERR-DC-002 (Weeping), ERR-DC-003 (Foaming)**
- ****Replacement_Part_Number:** TSI-DC-FRAC-001-REV-C**
- ****Maintenance_Schedule:** Quarterly inspections; Annual internal cleaning; 5-year major overhaul**

SECTION 1 – USE CASE

- Separates crude oil into light naphtha, heavy naphtha, kerosene, and gas oil fractions via temperature gradient
- Operates continuously at 350–380°C overhead; 380–400°C at reboiler

- Processes 8,500–10,000 barrels per day (BPD) feedstock
- Maintains 40–50 theoretical stages with bubble-cap trays for efficient vapor-liquid contact
- Feeds downstream hydrotreating and reforming units

SECTION 2 – FAILURE CAUSES

- **ERR-DC-001 (Tray Flooding):** Excessive vapor velocity; occurs 35% of incidents; causes product quality degradation
- **ERR-DC-002 (Weeping):** Low liquid flow rates; 28% frequency; results in incomplete separation
- **ERR-DC-003 (Foaming):** Crude contaminants (salts, water); 22% frequency; reduces tray efficiency
- **Corrosion/Erosion:** High-temperature sulfidic crude; 10% frequency; affects tray internals
- **Reboiler Tube Fouling:** Coke formation; 5% frequency; reduces heat transfer efficiency

SECTION 3 – SOLUTIONS

- **Tray Flooding:** Reduce feed rate 5–10%; inspect weir heights per API RP 54 Section 8.2; verify vapor distribution
- **Weeping Prevention:** Maintain minimum liquid flow; install anti-weep baffles; monitor differential pressure across trays
- **Foam Control:** Inject silicone antifoam agent (0.5–2 ppm); pre-treat crude with desalter per OSHA 1910.119 PSM
- **Corrosion Mitigation:** Use 316L stainless steel trays in high-temperature zones; conduct ultrasonic thickness testing annually
- **Reboiler Maintenance:** Chemical cleaning every 6 months; mechanical tube cleaning annually; replace fouled tubes per TSI specifications

SECTION 4 – INVENTORY DATA

- **Current Stock:** 1 unit operational; 0 spare units in inventory
- **Reorder Threshold:** Initiate procurement when operational unit reaches 85% tray damage threshold
- **Supplier:** Thermal Separation Industries Inc.; 18-week lead time; cost ~\$2.4M
- **Emergency Spare:** Recommend maintaining 1 spare unit; estimated storage cost \$45K/year

UNIT 2: OVERHEAD CONDENSER SYSTEM

Unit_Name: Air-Cooled Condenser (ACC-7A)

Unit_Id: DC-COND-002

Manufacture_Date: 2023-05-10

Manufacturer_Company_Name: Apex Thermal Equipment Corp.

Manufacturer_Company_Id: APEX-THERM-7834

Inventory_Quantity: 1

****Supplier_Information:**** Apex Thermal Direct; Dallas TX; Contact: sales@apexthermal.com; Lead Time: 14 weeks

DOMAIN SPECIFICATIONS

- ****Wellbore_Depth_Rating:**** N/A
- ****Operating_Pressure_PSI:**** 25–30 PSI (condensing light naphtha vapors)
- ****Drilling_Fluid_Compatibility:**** N/A (Hydrocarbon vapors: C5–C7 range)
- ****Known_Error_Codes:**** ERR-ACC-001 (Tube fouling), ERR-ACC-002 (Fan failure), ERR-ACC-003 (Pressure drop)
- ****Replacement_Part_Number:**** APEX-ACC-7A-STD
- ****Maintenance_Schedule:**** Monthly fan inspection; Quarterly tube cleaning; Semi-annual bearing replacement

SECTION 1 – USE CASE

- Condenses overhead vapors (350°C) from distillation column to liquid naphtha (~40°C)
- Air-cooled design with 12 tube bundles; 8 electric fans (0.75 kW each)
- Handles 2,500–3,200 BPD condensate flow
- Integrates with reflux drum for product separation and recycle control
- Critical for maintaining column pressure and separation efficiency

SECTION 2 – FAILURE CAUSES

- ****ERR-ACC-001 (Tube Fouling):**** Coke/gum deposition; 42% frequency; reduces cooling capacity by 15–25%
- ****ERR-ACC-002 (Fan Motor Failure):**** Bearing wear, electrical overload; 28% frequency; loss of cooling capacity
- ****ERR-ACC-003 (Pressure Drop):**** Tube blockage, fin corrosion; 18% frequency; increases column pressure
- ****Vibration/Noise:**** Loose tube supports, fan imbalance; 8% frequency; structural damage risk
- ****Seal Leakage:**** Gasket degradation; 4% frequency; hydrocarbon vapor release

SECTION 3 – SOLUTIONS

- ****Tube Fouling:**** Perform chemical cleaning (solvent flush) quarterly; inspect tubes via borescope; replace corroded tubes per ASME B16.34
- ****Fan Maintenance:**** Replace motor bearings every 6 months; balance fan blades annually; monitor vibration <0.3 in/sec per ISO 10816
- ****Pressure Management:**** Install differential pressure gauge; clean tubes when ΔP exceeds 8 PSI; verify fin integrity
- ****Vibration Control:**** Tighten tube supports; inspect mounting bolts monthly; replace worn elastomeric isolators per OSHA 1910.212
- ****Seal Integrity:**** Replace gaskets annually; use PTFE-lined seals for hydrocarbon service; pressure-test after replacement

SECTION 4 – INVENTORY DATA

- **Current Stock:** 1 unit operational; 2 spare tube bundles in warehouse
- **Reorder Threshold:** Order replacement tubes when fouling exceeds 20% surface area
- **Supplier:** Apex Thermal Equipment Corp.; 14-week lead time; tube bundle cost ~\$185K
- **Spare Parts Kit:** Maintain 4 fan motors, 8 bearing sets, 12 gasket kits; annual cost ~\$32K

UNIT 3: REBOILER FURNACE SYSTEM

Unit_Name: Fired Reboiler Heater

Unit_Id: DC-REBR-003

Manufacture_Date: 2023-03-18

Manufacturer_Company_Name: Combustion Engineering Solutions Ltd.

Manufacturer_Company_Id: CES-COMB-5612

Inventory_Quantity: 1

Supplier_Information: CES Industrial Supply; New Orleans LA; Contact: tech@cesolutions.com;
Lead Time: 16 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** N/A
- **Operating_Pressure_PSI:** 40–50 PSI (reboiler duty; tube-side crude oil)
- **Drilling_Fluid_Compatibility:** N/A (Natural gas fuel; crude oil heating medium)
- **Known_Error_Codes:** ERR-REB-001 (Tube rupture), ERR-REB-002 (Burner misfire), ERR-REB-003 (Scale buildup)
- **Replacement_Part_Number:** CES-REBR-FIRED-STD-2023
- **Maintenance_Schedule:** Daily burner inspection; Weekly tube inspection; Monthly combustion analysis; Annual hydrostatic test

SECTION 1 – USE CASE

- Provides sensible and latent heat to vaporize crude oil bottoms for column reboiling
- Fired heater with 2 burners (5 MMBtu/hr each); natural gas fuel; 85% thermal efficiency
- Heats crude from 150°C inlet to 400°C outlet; duty ~10 MMBtu/hr
- Maintains column bottom temperature and vapor generation rate
- Integrates with fuel gas system and flue gas stack per API RP 54 Section 5

SECTION 2 – FAILURE CAUSES

- **ERR-REB-001 (Tube Rupture):** Overpressure, corrosion, thermal stress; 38% frequency; catastrophic failure risk
- **ERR-REB-002 (Burner Misfire):** Fuel supply interruption, ignition failure; 32% frequency; loss of heat input

- **ERR-REB-003 (Scale Buildup):** Mineral deposits on tube ID; 18% frequency; reduces heat transfer 10–20%
- **Refractory Damage:** Thermal cycling, mechanical impact; 8% frequency; heat loss, safety hazard
- **Flame Impingement:** Burner misalignment; 4% frequency; localized tube overheating

SECTION 3 – SOLUTIONS

- **Tube Integrity:** Conduct ultrasonic thickness testing annually; replace tubes showing <0.125" wall thickness; perform hydrostatic test at 1.5× operating pressure per ASME B31.3
- **Burner Management:** Inspect burner tips daily; verify fuel pressure 5–8 PSI; test ignition system weekly; clean fuel strainers monthly
- **Scale Prevention:** Inject scale inhibitor (0.5–1.0 ppm) into feed; perform chemical tube cleaning annually; monitor outlet temperature differential
- **Refractory Care:** Inspect for cracks monthly; repair spalls immediately; replace refractory every 5 years or per manufacturer guidance
- **Flame Monitoring:** Install UV flame scanner; verify flame pattern daily; adjust burner air/fuel ratio per combustion analysis; maintain O₂ 2–3% in flue gas

SECTION 4 – INVENTORY DATA

- **Current Stock:** 1 unit operational; 0 spare reboilers in inventory
- **Reorder Threshold:** Initiate procurement if tube replacement exceeds 15% of bundle
- **Supplier:** Combustion Engineering Solutions Ltd.; 16-week lead time; cost ~\$1.8M
- **Spare Parts:** Maintain 4 burner assemblies, 8 ignition transformers, 20 tube sections; annual cost ~\$48K

UNIT 4: REFLUX DRUM & PRODUCT SEPARATION VESSEL

Unit_Name: Overhead Reflux Accumulator

Unit_Id: DC-REFL-004

Manufacture_Date: 2023-05-28

Manufacturer_Company_Name: Pressure Vessel Dynamics Inc.

Manufacturer_Company_Id: PVD-VESSEL-3847

Inventory_Quantity: 1

Supplier_Information: PVD Industrial; Houston TX; Contact: orders@pvdynamics.com; Lead Time: 12 weeks

DOMAIN SPECIFICATIONS

- **Wellbore_Depth_Rating:** N/A
- **Operating_Pressure_PSI:** 20–28 PSI (atmospheric + slight vacuum)
- **Drilling_Fluid_Compatibility:** N/A (Light naphtha liquid/vapor; C₅–C₇ hydrocarbons)

- **Known_Error_Codes:** ERR-RFD-001 (Level control failure), ERR-RFD-002 (Vapor carryover), ERR-RFD-003 (Corrosion)
- **Replacement_Part_Number:** PVD-REFL-DRUM-STD-500BBL
- **Maintenance_Schedule:** Daily level gauge check; Weekly drain valve operation; Quarterly internal inspection; Annual corrosion coupon analysis

SECTION 1 – USE CASE

- Accumulates condensed overhead vapors; separates liquid reflux from product naphtha
- 500-barrel capacity; horizontal orientation; internal demister pad for vapor separation
- Maintains reflux flow to column (typically 60–80% of overhead); product withdrawal ~20–40%
- Integrates level control system with column feed/reflux ratio optimization
- Critical for column stability and product quality control

SECTION 2 – FAILURE CAUSES

- **ERR-RFD-001 (Level Control Failure):** Float valve stiction, instrument drift; 40% frequency; causes column upset
- **ERR-RFD-002 (Vapor Carryover):** Demister pad fouling, high vapor velocity; 35% frequency; liquid entrainment to downstream units
- **ERR-RFD-003 (Corrosion):** Water ingress, sulfidic crude; 15% frequency; pitting, perforation risk
- **Drain Valve Blockage:** Coke/gum deposits; 7% frequency; prevents sump drainage
- **Gasket Leakage:** Thermal cycling, mechanical vibration; 3% frequency; hydrocarbon vapor release

SECTION 3 – SOLUTIONS

- **Level Control:** Calibrate float valve quarterly; replace stiction-prone seals with PTFE-lined designs; install redundant level transmitter per OSHA 1910.119
- **Demister Maintenance:** Inspect pad monthly; clean with solvent spray if fouling >10%; replace pad every 2 years or per manufacturer specification
- **Corrosion Management:** Conduct annual internal inspection via borescope; install corrosion coupons; apply internal epoxy coating if pitting detected; drain water daily
- **Drain System:** Operate drain valve weekly to prevent blockage; install strainer upstream of valve; flush with hot oil if clogged
- **Seal Integrity:** Replace gaskets annually; use spiral-wound gaskets with PTFE filler for hydrocarbon service; torque bolts per ASME B16.5 specifications

SECTION 4 – INVENTORY DATA

- **Current Stock:** 1 unit operational; 0 spare drums in inventory
- **Reorder Threshold:** Order replacement if internal corrosion exceeds 0.050" depth or demister pad replacement exceeds 3 cycles/year
- **Supplier:** Pressure Vessel Dynamics Inc.; 12-week lead time; cost ~\$420K
- **Spare Parts:** Maintain 2 float valve assemblies, 4 demister pads, 12 gasket sets, 6 drain valve cartridges; annual cost ~\$18K

COMPLIANCE & SAFETY SUMMARY

Regulatory Alignment:

- All procedures comply with OSHA 1910.119 (Process Safety Management)
- Equipment design meets API RP 54 (Refinery Safety) standards
- Pressure vessels certified per ASME Section VIII Division 1
- Maintenance intervals follow API RP 571 (Damage Mechanisms)

Emergency Contacts:

- Field Engineer: James R. Hutch